



AQ8.5: Activity Questions 5 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Choose the correct options.

☐

Let $T: R^2 \rightarrow R^2$ be a linear transformation, where R^2 is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.

☐

Let $T: R^2 \rightarrow R^2$ be an orthogonal linear transformation, where R^2 is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.

☐

Let $T: R^2 \rightarrow R^2$ be a linear transformation, where R^2 is the inner product space with respect to the inner product given by $\langle a, b \rangle = 2a_1 b_1 + 5a_2 b_2$. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$.

☐

Let $T: R^2 \rightarrow R^2$ be an orthogonal linear transformation, where R^2 is the inner product space with respect to the inner product given by $\langle a, b \rangle = 2a_1 b_1 + 5a_2 b_2$. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let $T: R^2 \rightarrow R^2$ be an orthogonal linear transformation, where R^2 is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.

Let $T: R^2 \rightarrow R^2$ be an orthogonal linear transformation, where R^2 is the inner product space with respect to the inner product given by $\langle a, b \rangle = 2a_1 b_1 + 5a_2 b_2$. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$.

1 point

If A is the matrix representation of an orthogonal transformation $T: R^n \rightarrow R^n$, then choose the set of correct statements.

☐

$$AA^T = I \quad A^T A = I$$

☐

$$A = A^T \quad A = A^T$$

☐

AA^T is an upper triangular matrix

☐

AA^T is a lower triangular matrix

☐

$$A^T = A^{-1} \quad A^T = A^{-1}$$

☐

$$A^T = -A^{-1} \quad A^T = -A^{-1}$$

☐

A^{-1} does not exist

☐

A^{-1} is also an orthogonal matrix

No, the answer is incorrect.

Score: 0



Accepted Answers:

$$AA^T = I \quad AAT = I$$

AA^T is an upper triangular matrix

AA^T is a lower triangular matrix

$$A^T = A^{-1} \quad AT = A^{-1}$$

$A^{-1}A^{-1}$ is also an orthogonal matrix

1 point

If AA is the matrix representation of an orthogonal transformation $T: R^n \rightarrow R^n$, then choose the set of correct statements.

☐

The column vectors of AA are orthogonal but not orthonormal.

☐

The column vectors of AA are orthonormal.

☐

The row vectors of AA are orthogonal but not orthonormal.

☐

The row vectors of AA are orthonormal.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The column vectors of AA are orthonormal.

The row vectors of AA are orthonormal.

Level 2

1 point

Consider the orthogonal set $B = \{(1, 1, 1), (-1, 1, 0)\}$ in R^3 . Find a third vector in R^3 which is orthogonal to both the vectors in B .

☐

$(1, 1, -2)$

☐

$(1, -1, 2)$

☐

$(-1, 1, -2)$

☐

$(-1, -1, -2)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, 1, -2)$

1 point

Consider the orthogonal set $B = \{(1, 1, 1), (-1, 1, 0)\}$ and vector obtained in question 4 which is orthogonal to the set B . Form the columns of an orthogonal matrix U by scaling these vectors appropriately. Choose the correct option.

☐

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & -2 & 0 \end{bmatrix}$$

☐

$$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{bmatrix} U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

A diagram illustrating a memory stack or array structure. It consists of eight horizontal bars stacked vertically. Each bar contains a mathematical expression involving a square root and a number. The expressions from top to bottom are:

- $\sqrt{13}$
- $\sqrt{13}$
- $\sqrt{12}$
- $\sqrt{-12}$
- $\sqrt{106}$
- $\sqrt{16}$
- $\sqrt{16}$
- $\sqrt{-2}$

A small circle is positioned at the bottom left corner of the diagram.

$$U = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix} U = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{bmatrix} U = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$$

$\sqrt{1}$	3
$\sqrt{1}$	3
$\sqrt{1}$	2
$\sqrt{-1}$	2
$\sqrt{10}$	6
$\sqrt{-1}$	6
$\sqrt{1}$	6
$\sqrt{1}$	

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix} U = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

$$\sqrt{10}$$

$$\sqrt{10}$$

$$\sqrt{10}$$

$$\sqrt{10}$$

$$\sqrt{10}$$

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$$\sqrt{10}$$

1 point

Consider a vector $\mathbf{v} = (3, 0, 4)$ and U from Question 5. Choose the correct option using the dot product as the standard inner product.

☐

$$\|\mathbf{v}\| = \|U\mathbf{v}\| \|\mathbf{v}\| = \|U\mathbf{v}\|$$

☐

$$\|\mathbf{v}\| \neq \|U\mathbf{v}\| \|\mathbf{v}\| = \|U\mathbf{v}\|$$

☐

$$2\|\mathbf{v}\| = \|U\mathbf{v}\| 2\|\mathbf{v}\| = \|U\mathbf{v}\|$$

☐

$$\|\mathbf{v}\| = 2\|U\mathbf{v}\| \|\mathbf{v}\| = 2\|U\mathbf{v}\|$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\|\mathbf{v}\| = \|U\mathbf{v}\| \|\mathbf{v}\| = \|U\mathbf{v}\|$$

1 point

Let the matrix representation of a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be $U = \begin{bmatrix} \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$ $U =$

$$\begin{bmatrix} \\ \\ 2 \end{bmatrix}$$

$$\sqrt{-1 \quad 2}$$

$$\sqrt{10 \quad 6}$$

$$\sqrt{-1 \quad 6}$$

$$\sqrt{-1 \quad 6}$$

$$\sqrt{2 \quad 3}$$

$$\sqrt{1 \quad 3}$$

$$\sqrt{1 \quad 3}$$

$$\sqrt{1}.$$

Let $B = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 1 \end{bmatrix}$. Choose the set of correct statements.

☐

T^T is an orthogonal transformation.

☐

T^T is not an orthogonal transformation.

☐

BB is a real symmetric matrix.

☐

$U^{-1}BUU^{-1}BU$ is a diagonal matrix.

☐

$U^{-1}BUU^{-1}BU$ is a scalar matrix.

☐

$U^{-1}BUU^{-1}BU$ is an upper triangular matrix.

☐

$U^{-1}BUU^{-1}BU$ is a lower triangular matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T^T is an orthogonal transformation.

BB is a real symmetric matrix.

$U^{-1}BUU^{-1}BU$ is a diagonal matrix